

INSTRUCTIONS FOR RUNNING THE JUPYTER NOTEBOOK AND STREAMLIT APP

The detailed steps for running the Jupyter Notebook and Streamlit App are explained below:

1. INSTRUCTION FOR RUNNING THE JUPYTER NOTE BOOK:

The folder contains two Jupyter note book files:

- i. **Chatbot_Open.ipynb**: Open-source model
- ii. **Chatbot_GPT.ipynb**: OpenAI model

Due to the faster response time and higher embedding dimensions of the OpenAI model, results from Chatbot_GPT.ipynb are considered. The steps to run this notebook are outlined below:

- a. Create a **.env** file and insert your OpenAI API key.
- b. Open the **Chatbot_GPT.ipynb** notebook and execute each cell sequentially.
- c. Detailed instructions are already included in the notebook itself.
- d. Run the notebook to generate responses.
- e. **Important:** The notebook uses the OpenAI embedding model to convert text chunks into vector embeddings. When varying chunk sizes and overlap settings, the embedding vectors change each time, increasing API costs. To mitigate this, the *Diversity-Based Chunking Algorithm* is used to calculate the optimal chunk size and overlap. Once the optimal chunking parameters are identified, embeddings are computed only once and stored.
- f. These final embeddings are saved locally in the file **“embedded_chunks_safe.pkl”** to avoid recalculating them each time, thus reducing cost and improving performance.
- g. Use this saved embedding file **“embedded_chunks_safe.pkl”** for further processing.

2. INSTRUCTION TO RUN STREAMLIT APP:

The Streamlit app allows to deploy and interact with the chatbot in real time. Follow the steps below to get it running:

- a. Create a virtual environment using the command:

py -m venv venv

- b. Activate the virtual environment:

.\venv\Scripts\activate

- c. Install all required packages from “requirements.txt” file:

```
pip install -r requirements.txt
pip install -U langchain-community
```

- d. Run the Streamlit App (**app_new.py**):

```
streamlit run app.py
```

- e. The app will launch in the default browser.

Local Q&A Chatbot with Citations

 Enter your OpenAI API Key:

 Upload your embedded `.pkl` file



Drag and drop file here

Limit 200MB per file • PKL

Browse files



Please enter your OpenAI API key.

- f. Enter **OpenAI_API_KEY** = ” ##### ”
- g. Upload the embedding file “*embedded_chunks_safe.pkl*”. This file contains precomputed embedding vectors using the OpenAI embedding model with optimized chunk settings. Uploading this file to avoids recalculating embeddings each time, saving on cost and improving speed.
- h. **Enter your query** in the input box and hit Enter.
- i. The chatbot will then generate and display a relevant response.